ESPaperPlay / DEVELOPMENT BOARD r6

ESP32-S3-WROOM-1-N16R8 / GDEY075T7-T01 / protected 1-cell battery

Ten functional sheets. Wired functional modules; r5 electrical design unchanged.

POWER_INPUT_CHARGER

Input / charger / battery

文件: POWER_INPUT_CHARGER.kicad_sch

POWER_3V3_CONTROL

Regulator / IO expander / gauge

文件: POWER_3V3_CONTROL.kicad_sch

MCU_CORE

N16R8 / reset / user keys

文件: MCU_CORE.kicad_sch

USB_INPUT

USB-C / protection / source select

文件: USB_INPUT.kicad_sch

POWER_DOMAINS

Five switched power domains

文件: POWER_DOMAINS.kicad_sch

TOUCH_ISOLATION

GT911 four-signal isolation

文件: TOUCH_ISOLATION.kicad_sch

EPD_DISPLAY

UC8179 / boost / 24-pin FPC

文件: EPD_DISPLAY.kicad_sch

STORAGE_SD

microSD / bus isolation / detect

文件: STORAGE_SD.kicad_sch

AUDIO

I2S amplifier / microphone

文件: AUDIO.kicad_sch

CHARGE_TEMP

NTC window / hardware TS pause

文件: CHARGE_TEMP.kicad_sch

ASSEMBLY / REVIEW BOUNDARIES

0.70A nominal; external NTC window uses an assumed Honeywell curve.
R108 stays DNP; J105 open. No charging or manufacturing release.
J502 and J202 each use ONE shunt. Actual pack R-T / thermal tests pending.
r6 redraw preserves r5 values, candidate parts, UUIDs and all pin connections.

hardware/devboard-r0/README.md / docs/v2/hardware/power-design-spec.md

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /
File: ESPaperPlay_Devboard.kicad_sch

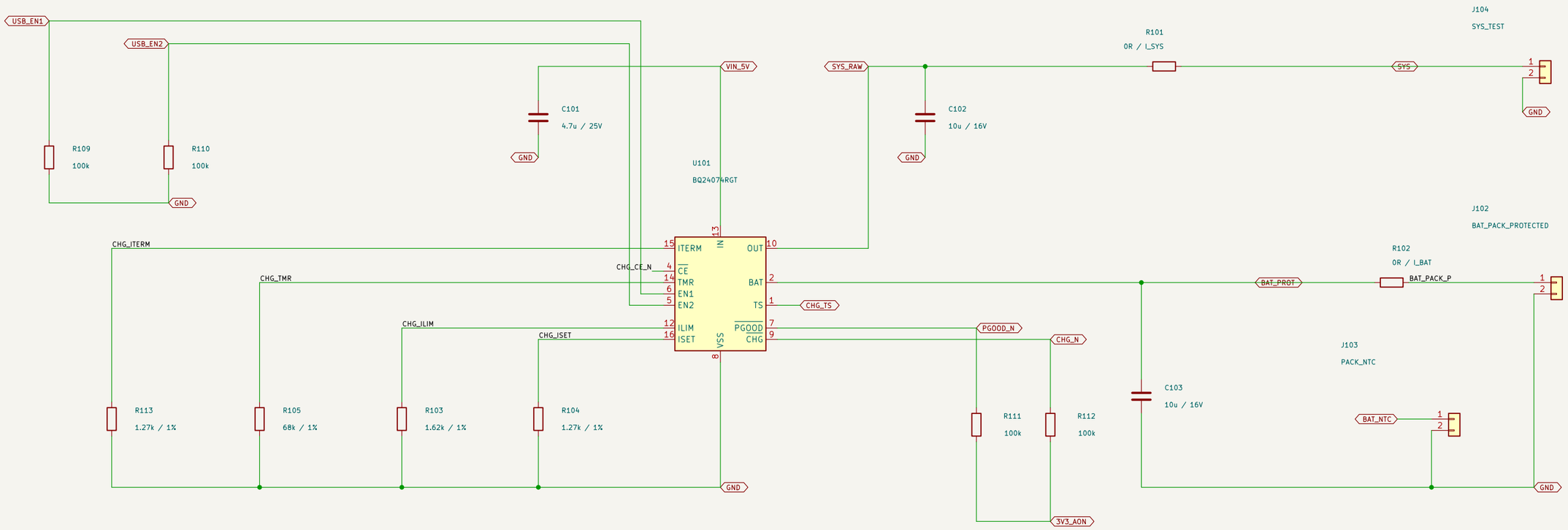
Title: ESPaperPlay DEVELOPMENT BOARD — PERIPHERAL DRAFT

Size: A3	Date: 2026-11-05	**Rev: r6 DRAWING**
KiCad E.D.A. 10.0.7		Id: 1/11

CHARGER / CONNECTED POWER PATH AND CONTROL MODULES

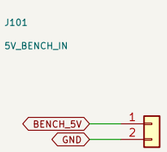
r6 drawing cleanup only. r5 component values, candidates, UUIDs and electrical connectivity are preserved.

1 BQ24074 / CURRENT SETTINGS / BATTERY AND SYSTEM PATH



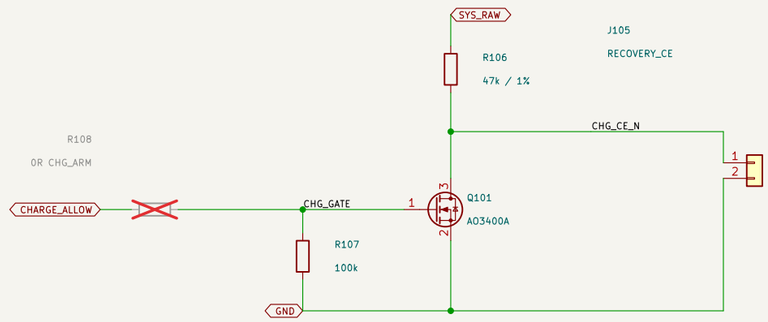
ISET/ITERM 1.27k: ~0.701A / 30mA nominal. TMR68k: ~9.07h. DPPM / heat can reduce charging current.

2 BENCH INPUT

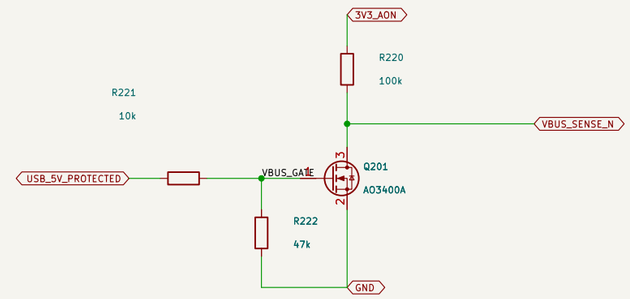


Known regulated,
current-limited 5V.
Select at J502;
do not bridge sources.

3 DEFAULT CHARGE INHIBIT / MAINTENANCE



4 PROTECTED-USB PRESENCE



VIN_5V GND SYS 3V3_AON

R108 remains DNP. J105 normally OPEN; maintenance CE does not bypass the page11 temperature window.
Default EN2/EN1=00 USB100. Raise input mode only for a known capable source; TS fault is verified by BAT current, not CHG_N alone.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /POWER_INPUT_CHARGER/
File: POWER_INPUT_CHARGER.kicad_sch

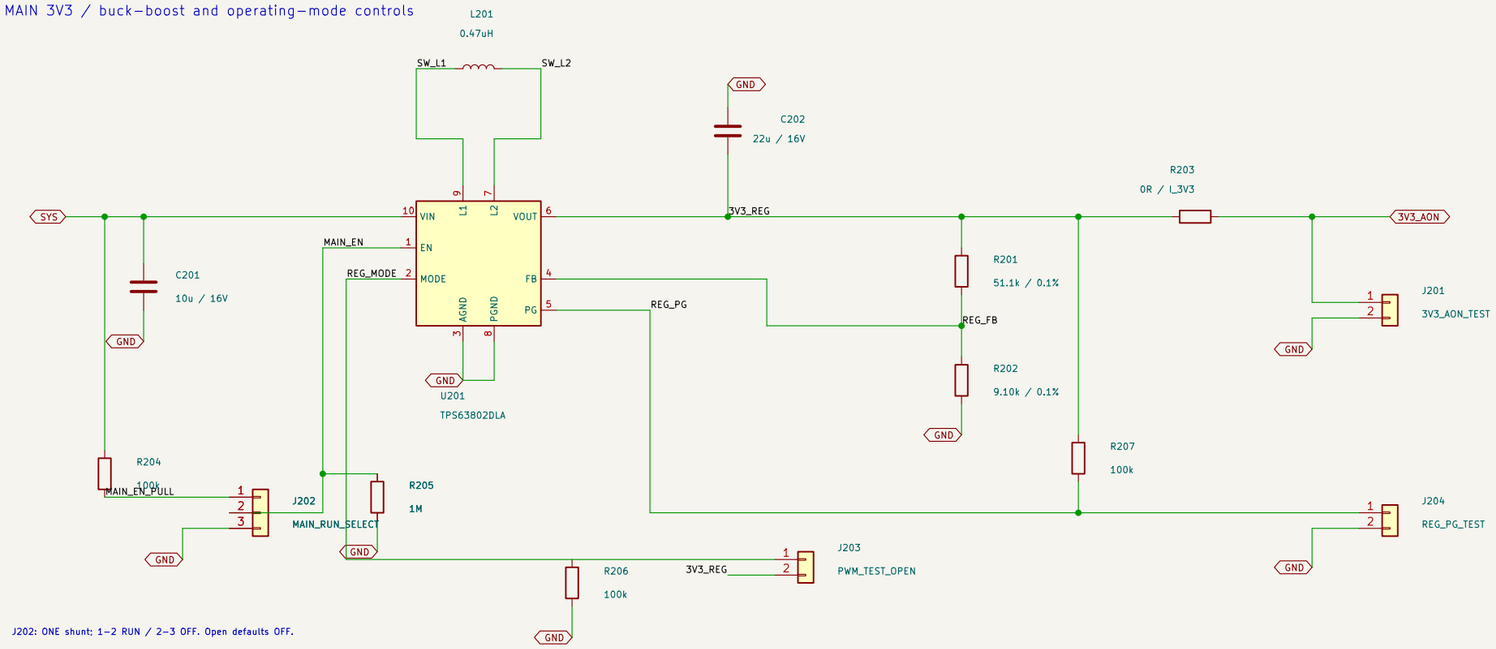
Title: POWER INPUT / BQ24074 CHARGER

Size: A3	Date: 2026-11-05	**Rev: r6 DRAWING**
KiCad E.D.A. 10.0.7		Id: 2/11

3V3 POWER / GPIO CONTROL / BATTERY GAUGE

Electrical design unchanged. Local support components are wired to their ICs; named ports cross module boundaries.

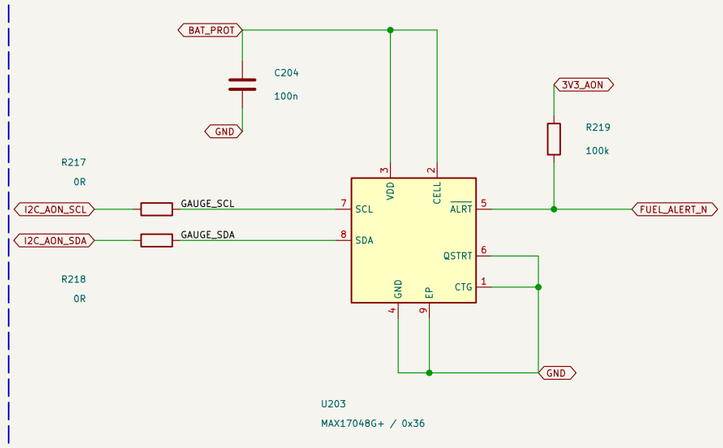
MAIN 3V3 / buck-boost and operating-mode controls



J202: ONE shunt; 1-2 RUN / 2-3 OFF. Open defaults OFF.

COUT / feedback stay before R203. PG is a test signal.

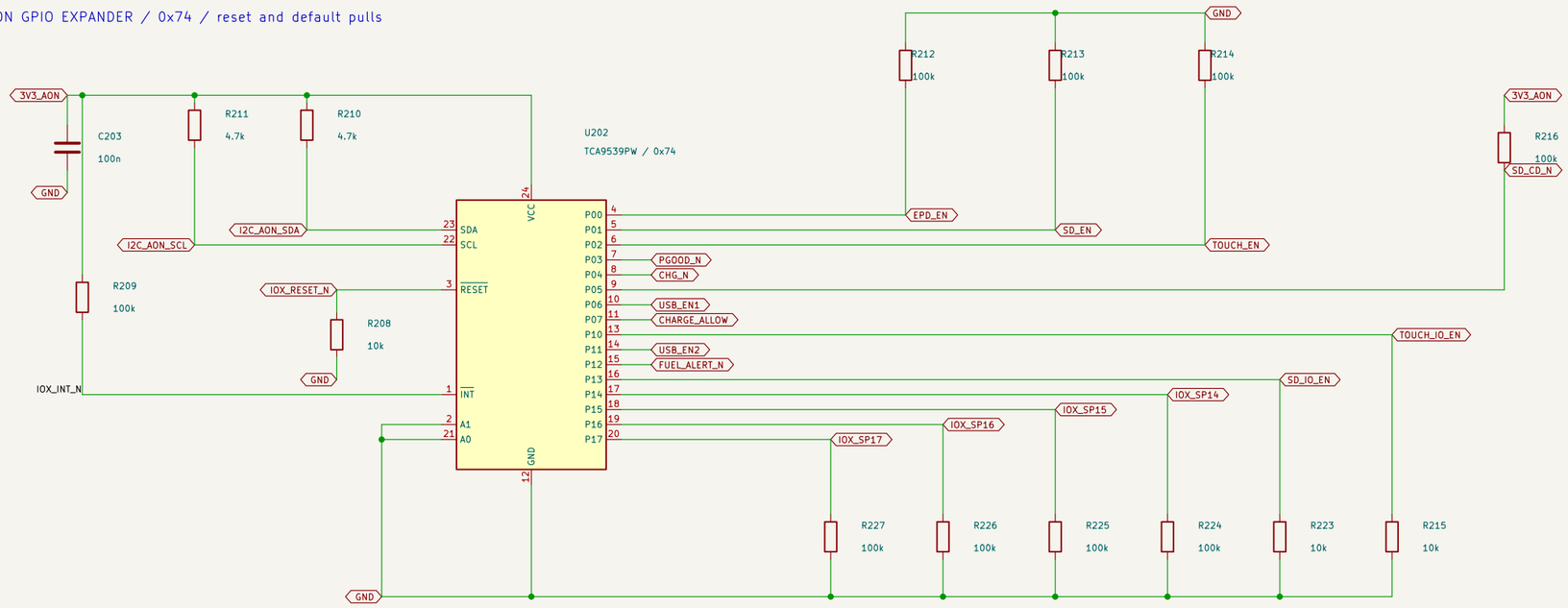
BATTERY GAUGE / 0x36



BAT-powered gauge: 0R links are debug disconnects.

Battery-absent Ioff / SOC accuracy still need tests.

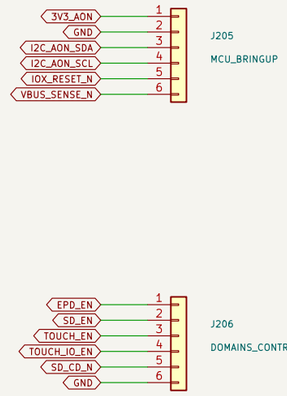
AON GPIO EXPANDER / 0x74 / reset and default pulls



RESET -> inputs; write safe latch BEFORE output direction. P14-P17 remain inputs.

P13 = SD_IO_EN. R212/R213/R214/R215/R223 provide the same hardware defaults.

BRING-UP / CONTROL PORTS



J205: MCU / I2C diagnostics

No external AON back-power.

J206: domain signals / GND

DESIGN REVIEW ONLY - NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /POWER_3V3_CONTROL/
File: POWER_3V3_CONTROL.kicad_sch

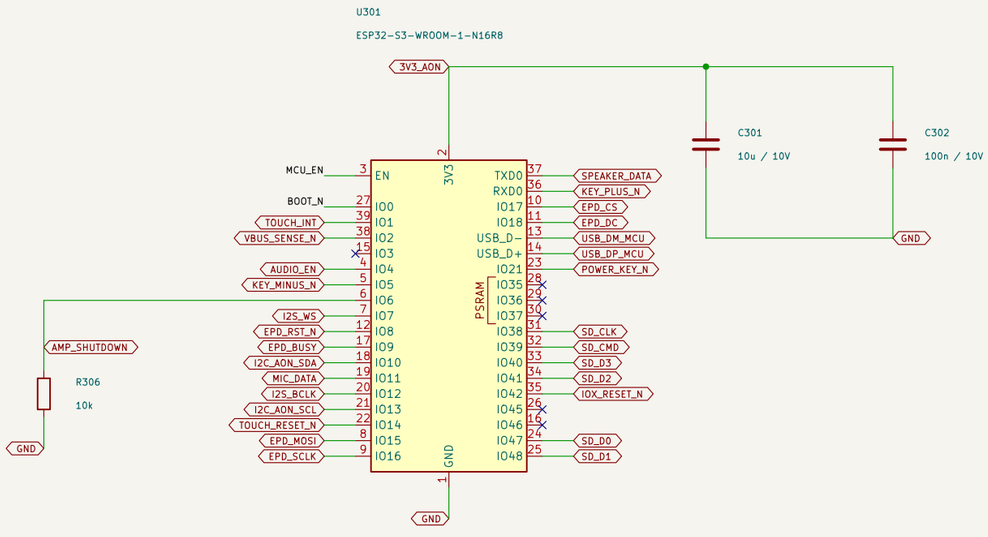
Title: 3.3 V REGULATOR / POWER CONTROL / GAUGE

Size: A3	Date: 2026-11-05	**Rev: r6 DRAWING**
KiCad E.D.A. 10.0.7		Id: 3/11

MCU CORE / CONNECTED FUNCTION BLOCKS

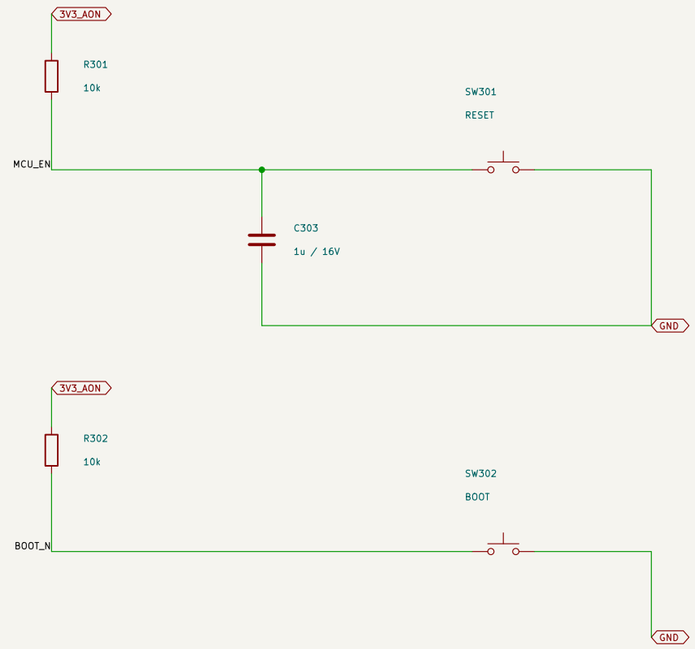
N16R8 physical pad mapping retained. Cross-page signals use named ports; local RC and key circuits are wired.

1 ESP32-S3 + LOCAL BYPASS / DEFAULT MUTE



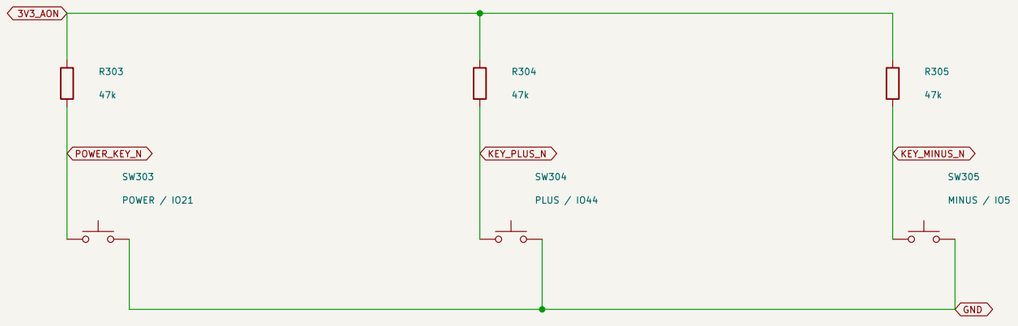
GPIO35/36/37 reserved for PSRAM. GPIO3/45/46 remain NC.
Module GND pads 1/40/41 include the exposed pad; preserve antenna keepout.

2 EN / BOOT MAINTENANCE



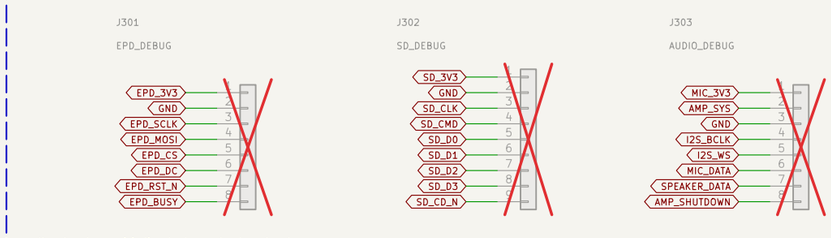
EN: 10k / 1uF; qualify supply ramp. BOOT: no large capacitor.
Hold BOOT and pulse RESET for ROM recovery; native USB is the default debug path.

3 USER KEYS / SOFTWARE DEBOUNCE



R306 holds the amplifier in shutdown at MCU reset. J301/J302/J303 remain DNP; values and connectivity unchanged.

4 DNP HIGH-IMPEDANCE DEBUG HEADERS



No external power injection; loads are already on board.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /MCU_CORE/
File: MCU_CORE.kicad_sch

Title: ESP32-S3 N16R8 / RESET / USER KEYS

Size: A3	Date: 2026-11-05	**Rev: r6 DRAWING**
KiCad E.D.A. 10.0.7		Id: 4/11

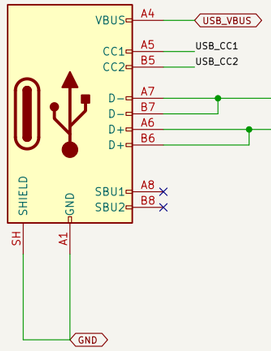
USB INPUT / CONNECTED PROTECTION AND DATA PATHS

USB device / 5V sink only. One source-select shunt. No PD, Host power or automatic CC current detection.

1 USB-C RECEPTACLE

J501

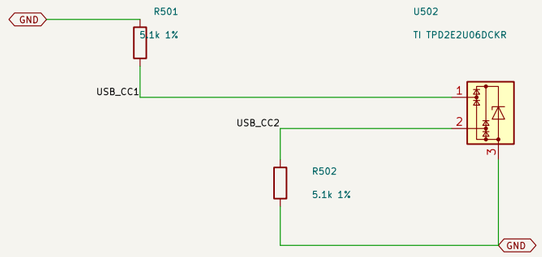
USB-C USB2 16P



USB_DP_PORT

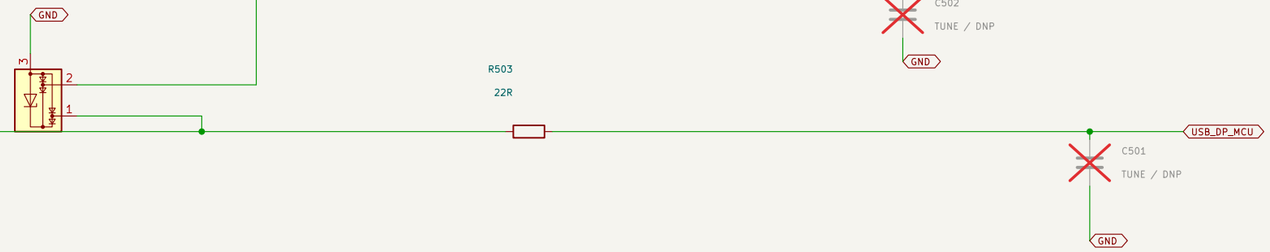
A6+B6 = D+
A7+B7 = D-
Shield to local GND.
SBU1/SBU2 NC.

2 CC PULL-DOWNS + ESD



3 DATA ESD + MCU-SIDE SERIES / TUNING

USB_DM_PORT



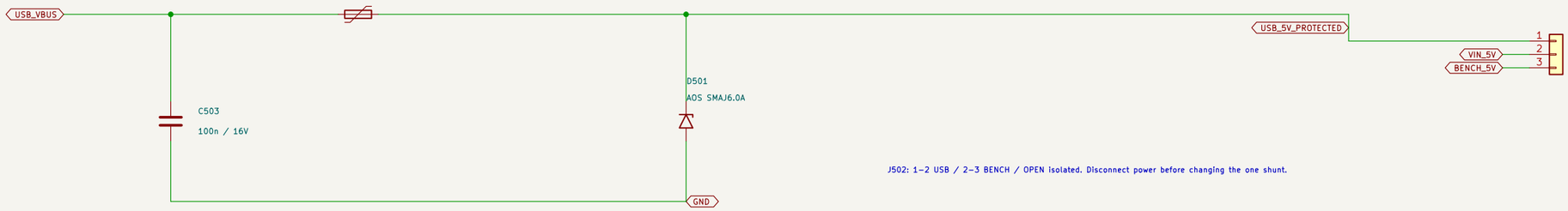
U501

TI TPD2E2U06DCKR

4 5V INPUT PROTECTION / MANUAL SOURCE SELECTION

F501

MF-MSMF200/16X-2



J502: 1-2 USB / 2-3 BENCH / OPEN isolated. Disconnect power before changing the one shunt.

PTC/TVS protect the USB branch only. Bench input must be known regulated, current-limited 5V.

D+/D-: short 90-ohm differential pair; 22R near MCU. C501/C502 remain DNP. CC resistors do not authorize >100mA.

DESIGN REVIEW ONLY - NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /USB_INPUT/

File: USB_INPUT.kicad_sch

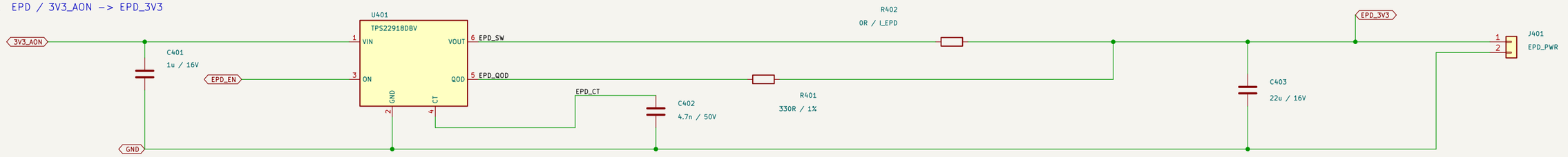
Title: USB-C DEVICE / ESD / 5V SOURCE SELECTION

Size: A3	Date: 2026-11-05	**Rev: r6 DRAWING**
KiCad E.D.A. 10.0.7		Id: 5/11

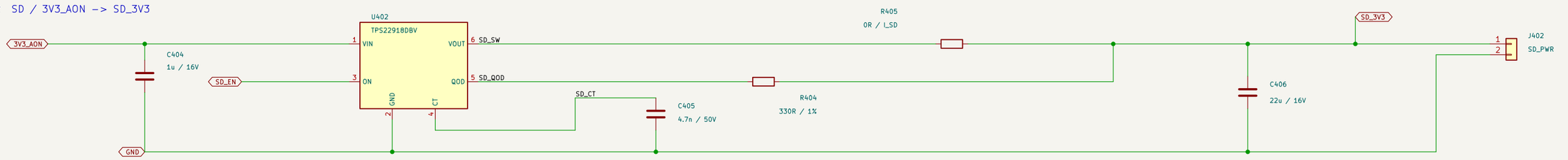
SWITCHED POWER DOMAINS / local bypass, timing and discharge wiring

OR links measure load current only. QOD returns to the LOAD side; remove both paths before external injection.

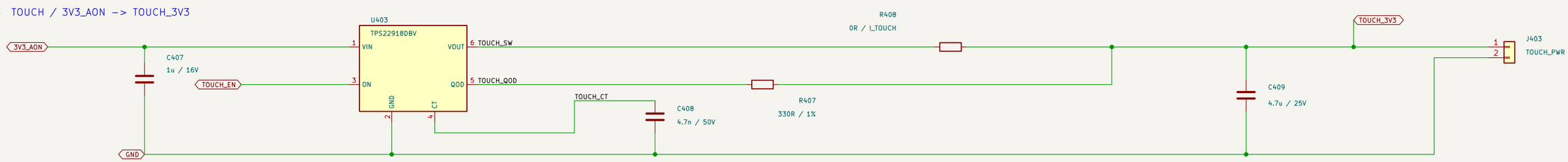
1 EPD / 3V3_AON -> EPD_3V3



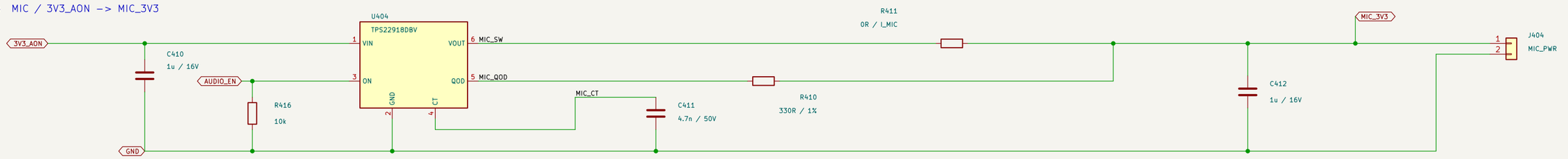
2 SD / 3V3_AON -> SD_3V3



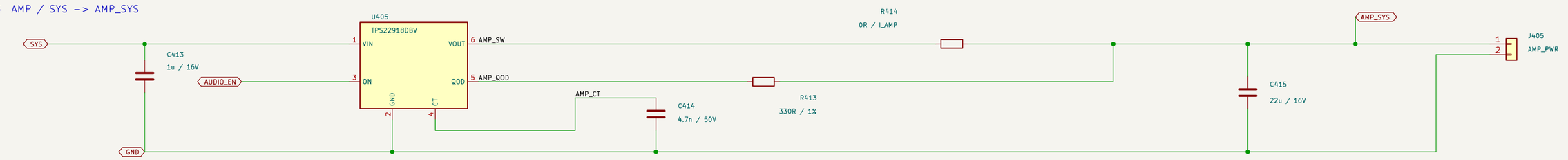
3 TOUCH / 3V3_AON -> TOUCH_3V3



4 MIC / 3V3_AON -> MIC_3V3



5 AMP / SYS -> AMP_SYS



Enable defaults: EPD / SD / TOUCH use R212 / R213 / R214 on the control page.

R416 holds MIC and AMP off together. SYS may remain alive with AON=0; verify AMP leakage.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /POWER_DOMAINS/

File: POWER_DOMAINS.kicad_sch

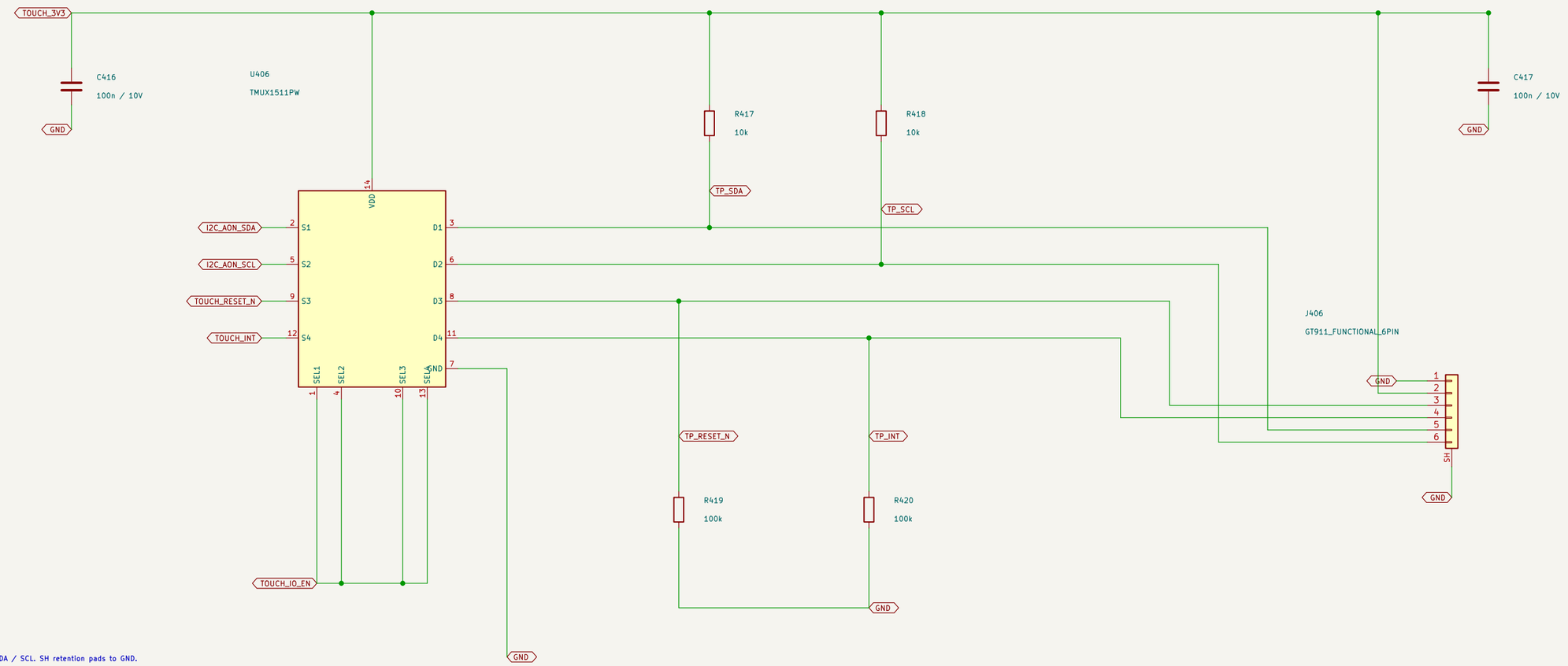
Title: PERIPHERAL POWER / FIVE SWITCHED DOMAINS

Size: A3	Date: 2026-11-05	**Rev: r6 DRAWING**
KiCad E.D.A. 10.0.7		Id: 6/11

TOUCH INTERFACE / DIRECT SIGNAL AND BIAS WIRING

TMUX1511 remains powered by TOUCH_3V3; MCU-side ports and all four signal isolation channels are unchanged.

1 DOMAIN BYPASS + FOUR-CHANNEL ISOLATION + PANEL INTERFACE



J406: GND / TOUCH_3V3 / RESET / INT / SDA / SCL. SH retention pads to GND.

2 POWER / RESET SEQUENCE AND ASSEMBLY NOTES

POWER ON: isolate signals -> enable TOUCH_3V3 -> qualify rail -> perform full GT911 address/reset sequence -> connect I/O.
POWER OFF: stop bus traffic -> disable touch wake -> isolate all four signals -> switch domain off.
R417/R418 are local 10k pull-ups; include the panel and AON pull-ups in rise-time/current checks. R419/R420 define RESET/INT defaults.
Retain the passive FP6 extension. Use the panel documentation for contact orientation/pin order; no active adapter function assumed.

TOUCH_3V3

The domain PWR_FLAG declares the intended powered state: it is not a protection component. Verify ramp, residual voltage and injection on the board.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

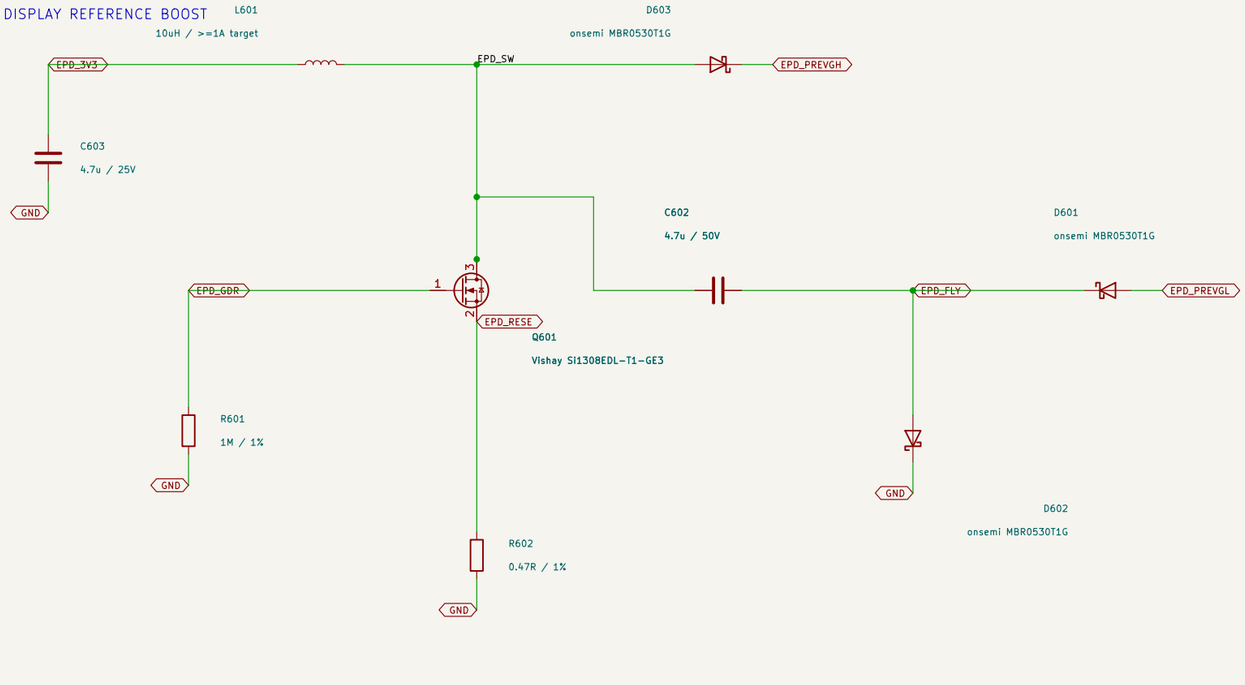
Sheet: /TOUCH_ISOLATION/
File: TOUCH_ISOLATION.kicad_sch

Title: GT911 FOUR-SIGNAL ISOLATION / DOMAIN TEST NOTES

Size: A3	Date: 2026-11-05	**Rev: r6 DRAWING**
KiCad E.D.A. 10.0.7		Id: 7/11

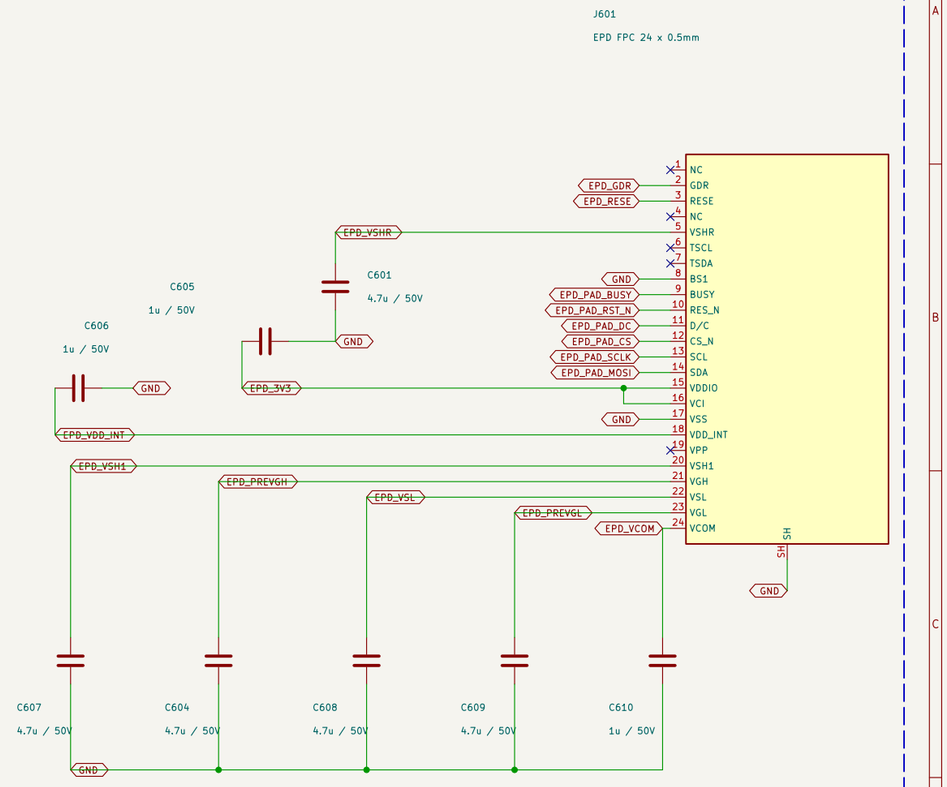
EPD / CONNECTED BOOST, PANEL STORAGE AND DOMAIN BUFFERS

GOOD DISPLAY REFERENCE BOOST

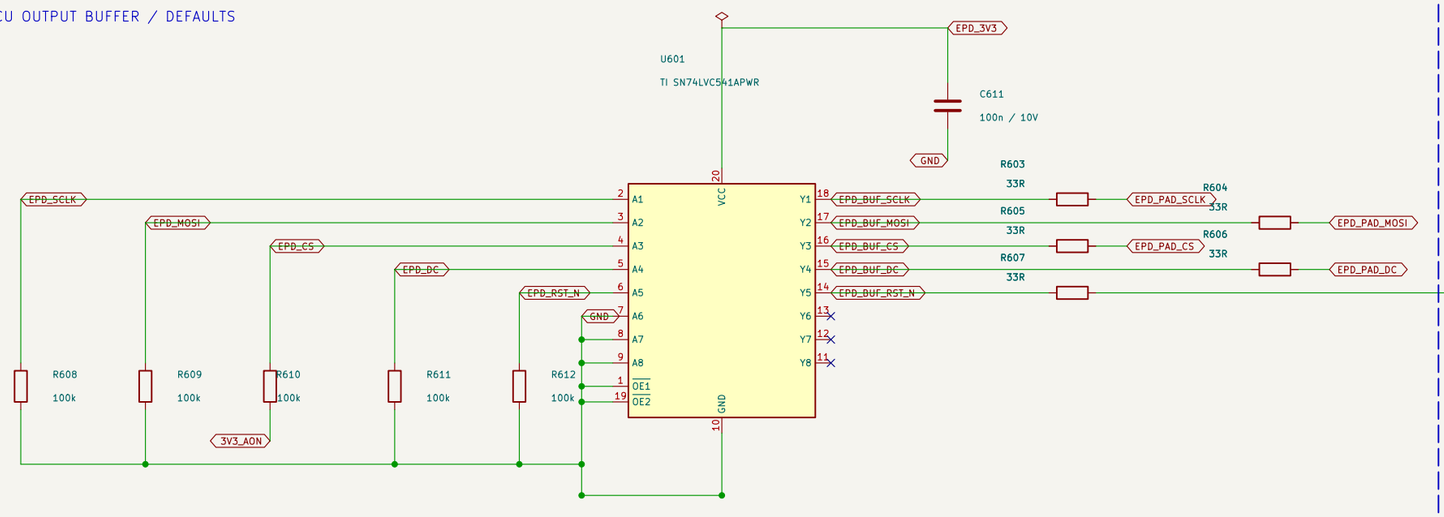


C602 is the flying capacitor: FLY to SW, never GND.
Q601 source -> RESE -> R602: keep switch loops short.
Panel generates HV. No external HV supply injection.

PANEL FPC / STORAGE CAPACITORS

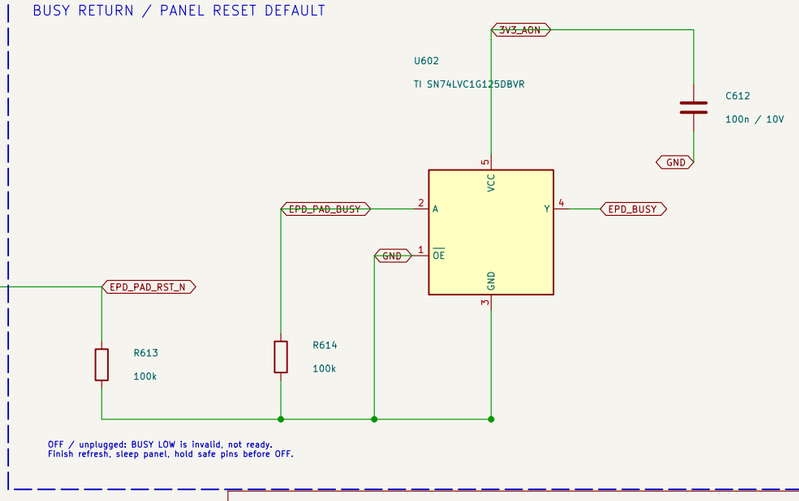


MCU OUTPUT BUFFER / DEFAULTS



CS defaults HIGH; CLK/MOSI/DC/RESET LOW. EPD rail supplies U601.

BUSY RETURN / PANEL RESET DEFAULT



OFF / unplugged: BUSY LOW is invalid, not ready.
Finish refresh, sleep panel, hold safe pins before OFF.

FPC uses retained passive extension. BS1=0, write-only SPI. VDD_INT has its own capacitor: it is not the 3.3V rail.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /EPD_DISPLAY/
File: EPD_DISPLAY.kicad_sch

Title: EPD REFERENCE BOOST / 24-PIN PANEL / DOMAIN BUFFERS

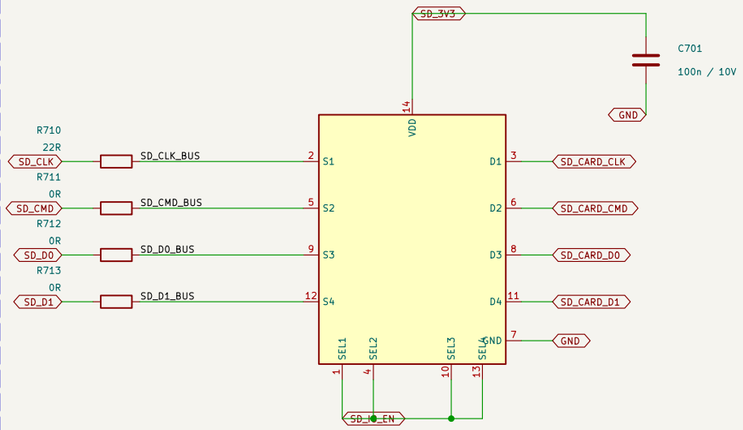
Size: A3	Date: 2026-11-05	**Rev: r6 DRAWING**
KiCad E.D.A. 10.0.7		Id: 8/11

MICROSD / WIRED ISOLATION, PULL-UPS AND CARD DETECT

CLK / CMD / D0 / D1

U701

TI TMUX1511PW

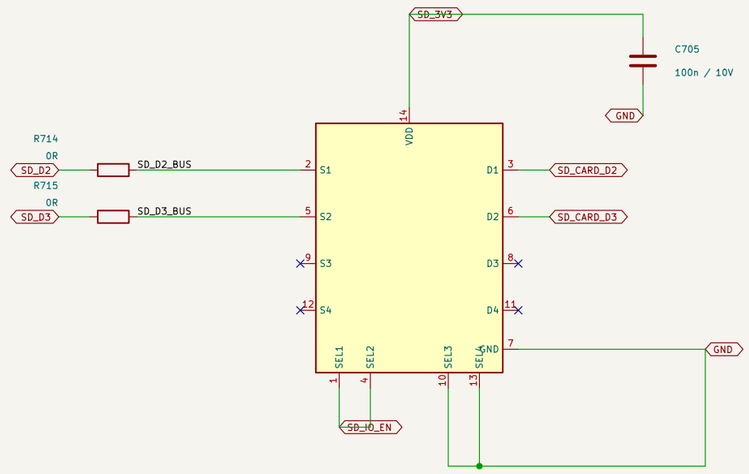


All SEL -> SD_IO_EN (IOX P13).
R223 default pull-down is on control page.

D2 / D3

U706

TI TMUX1511PW



Spare channels: SEL LOW, S/D NC.

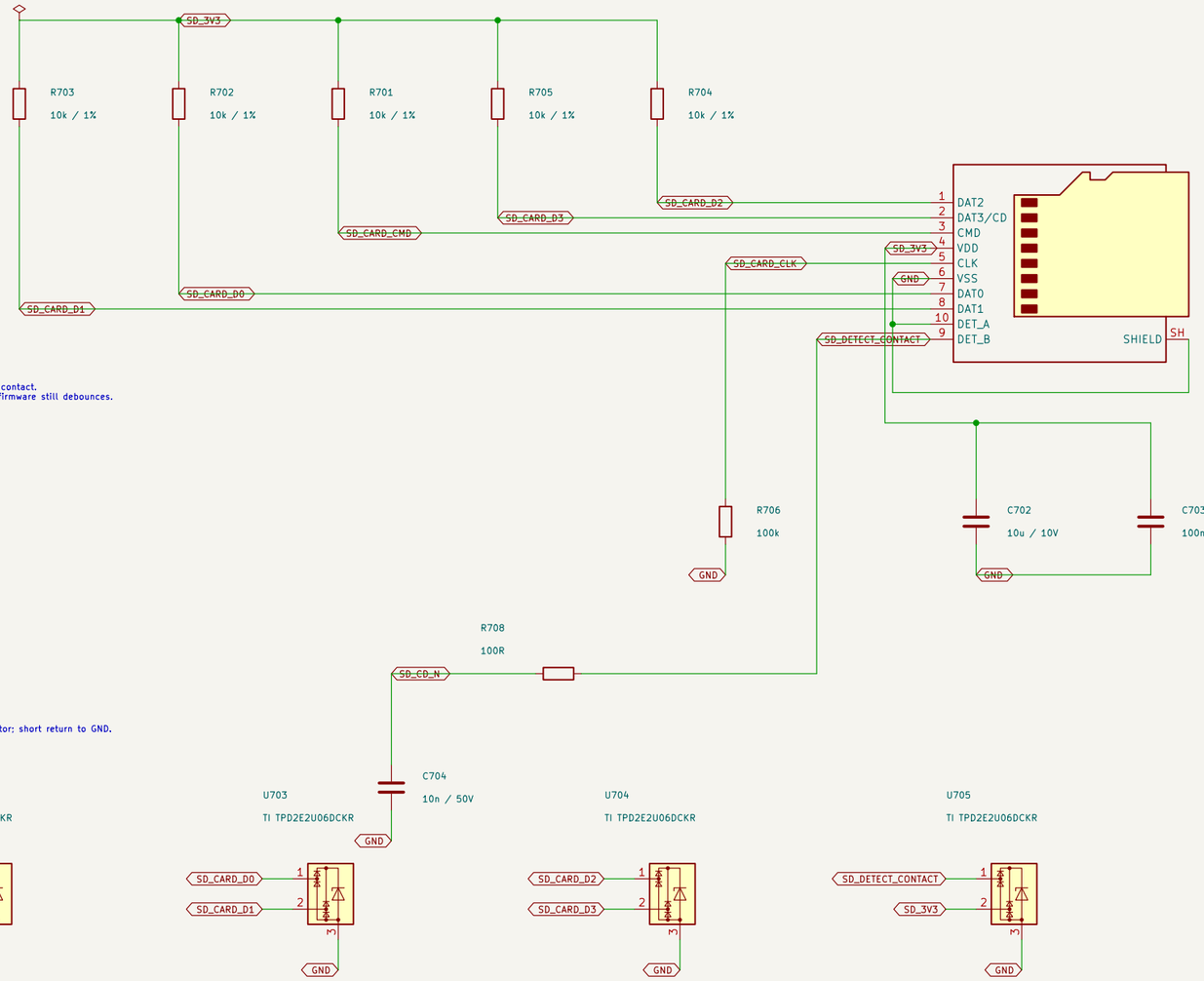
ON: isolate -> power valid -> connect I/O -> initialize. OFF: flush/unmount -> isolate -> power OFF.
No UHS 1.8V; begin at 400kHz init, then 4-bit 20MHz target. Values and connectivity unchanged from r5.

CARD SOCKET / PULL-UPS / DETECT / ESD

Five pull-ups use SD_3V3: CLK is pulled LOW.
Card power and SD_IO_EN are independently controlled.

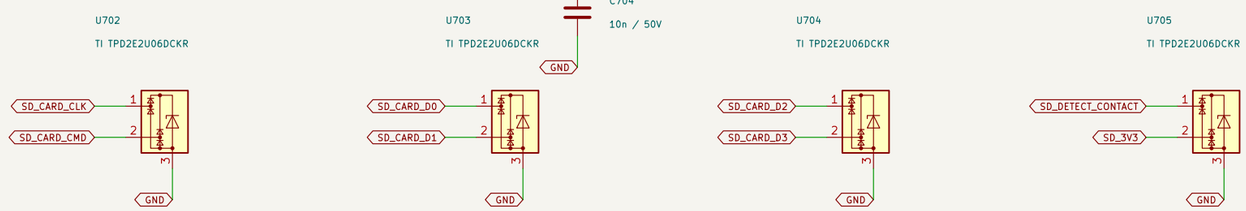
J701

Hirose DM3AT-SF-PEJM5



DAT3 is not the mechanical detect contact.
R708/C704 filter contact bounce; firmware still debounces.

Parallel ESD clamps at the connector; short return to GND.



DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /STORAGE_SD/

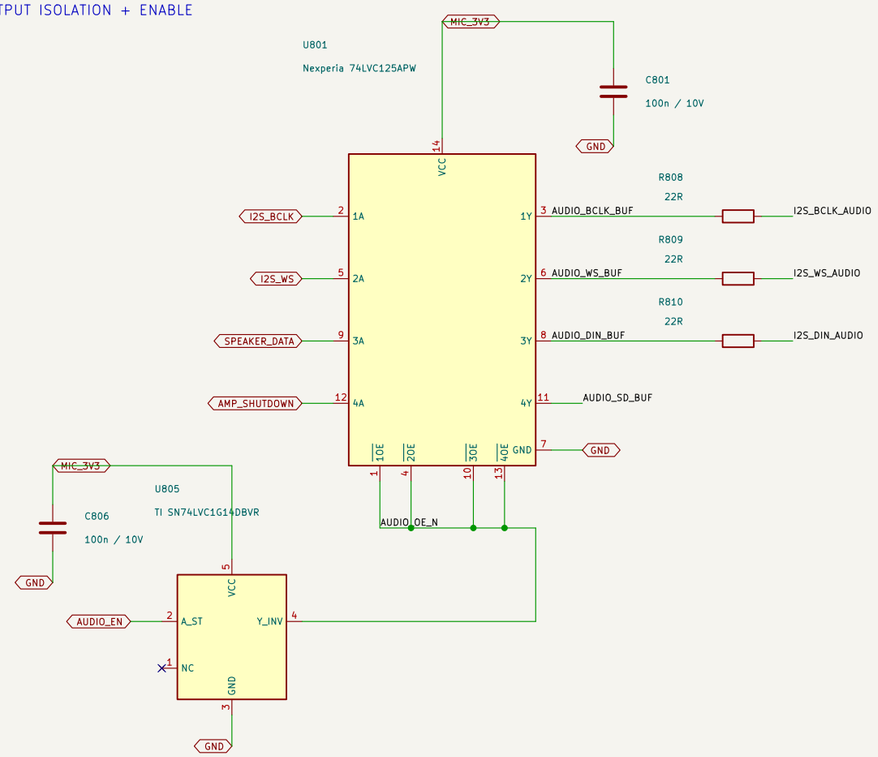
File: STORAGE_SD.kicad_sch

Title: MICROSD / 4-BIT BUS / INDEPENDENT SIGNAL ISOLATION

Size: A3	Date: 2026-11-05	**Rev: r6 DRAWING**
KiCad E.D.A. 10.0.7		Id: 9/11

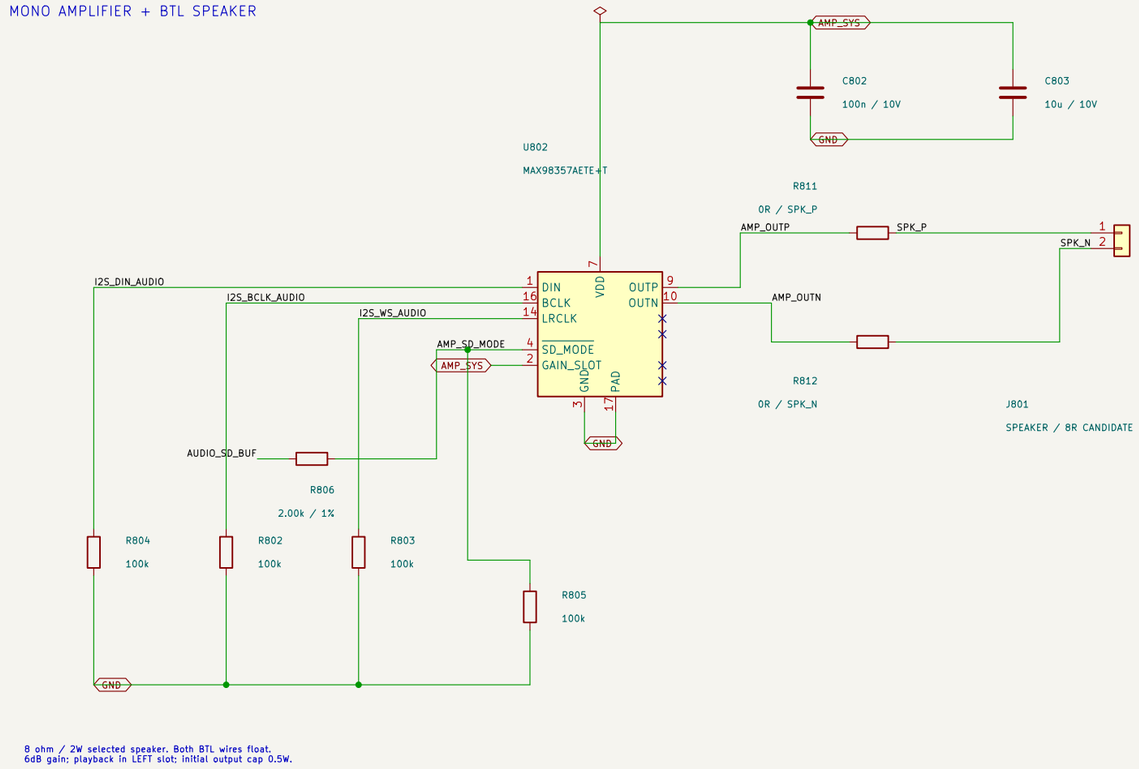
AUDIO / CONNECTED FUNCTIONAL MODULES

OUTPUT ISOLATION + ENABLE



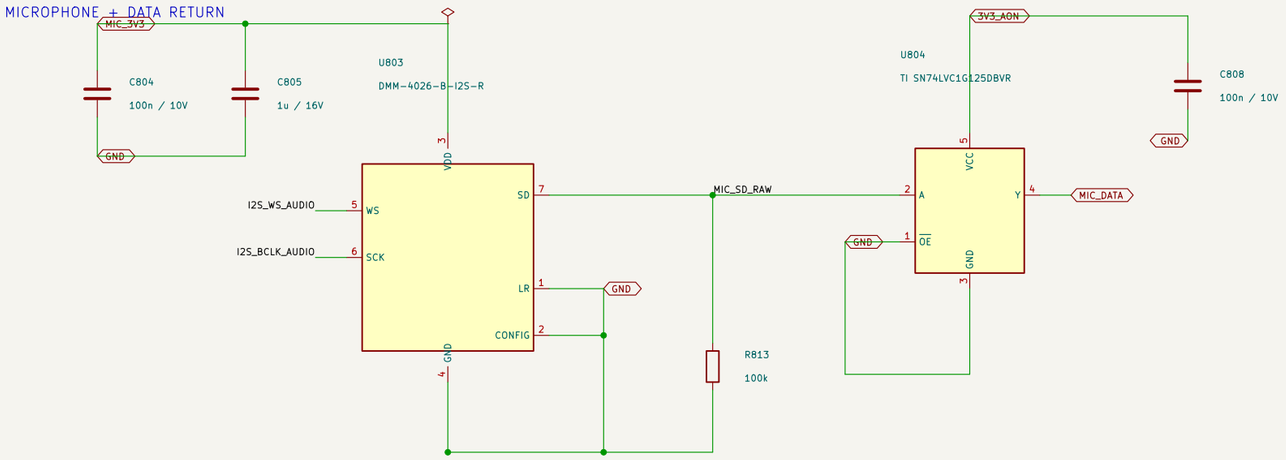
AUDIO_EN drives all four /OE pins.
U801/U805 powered from MIC_3V3; exact Ioff-rated vendor.

MONO AMPLIFIER + BTL SPEAKER



8 ohm / 2W selected speaker. Both BTL wires float.
6dB gain; playback in LEFT slot; initial output cap 0.5W.

MICROPHONE + DATA RETURN



LR and CONFIG strapped LOW. SD pull-down before AON return buffer.

OPERATING NOTES

48kHz / 2 x 32-bit I2S slots / 64fs.
Mute -> enable rails -> clocks/zero data
-> discard startup frames -> unmute.

OFF: mute, finish audio, stop clocks,
drive safe levels, then AUDIO_EN=0.
Verify low-battery THD, pop, rail decay,
SD_MODE injection and acoustic port.

No 5V audio boost; no new firmware.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /AUDIO/
File: AUDIO.kicad_sch

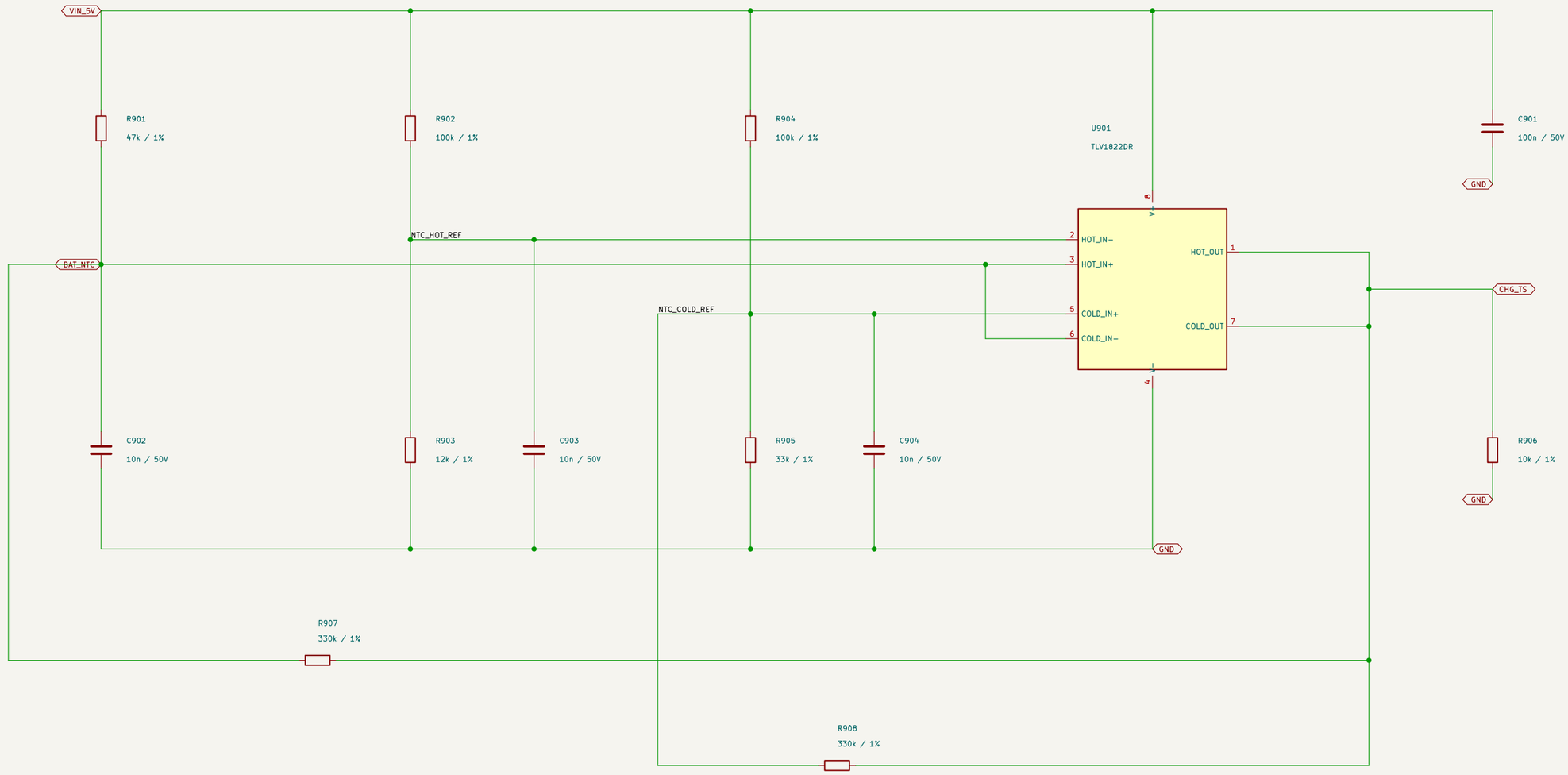
Title: AUDIO / I2S INPUT ISOLATION / MONO PLAYBACK + MIC

Size: A3	Date: 2026-11-05	**Rev: r6 DRAWING**
KiCad E.D.A. 10.0.7		Id: 10/11

CHARGE TEMPERATURE WINDOW / CONNECTED ANALOG CIRCUIT

Assumed Honeywell175-103LAE-301 Rev.B curve; actual pack NTC identity remains unverified. No electrical change from r5.

1 NTC SENSE / COLD AND HOT REFERENCES / OPEN-DRAIN WINDOW



330k feedback paths produce hysteresis; solve the three coupled nodes, not separate unloaded dividers.

5V nominal pause points: cold14.8C / hot40.1C. Resume example15.5C /39.5C assumes fault TS=0.1V.
NTC open -> cold fault; short -> hot fault. Either output pulls TS low, pausing charge and its timer.
U901 uses the same VIN as BQ24074. No battery-only divider load; independent comparator failure is not covered.
5V comparator offset/VOL are engineering budgets. Verify TS clamp, startup, hysteresis and actual pack R-T on hardware.
R108 stays DNP; J105 open. Reference review-r5/temperature for the unchanged circuit model and limitations.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

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Sheet: /CHARGE_TEMP/
File: CHARGE_TEMP.kicad_sch

Title: Battery temperature window / TS pause

Size: A3	Date: 2026-11-05	**Rev: r6 DRAWING**
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